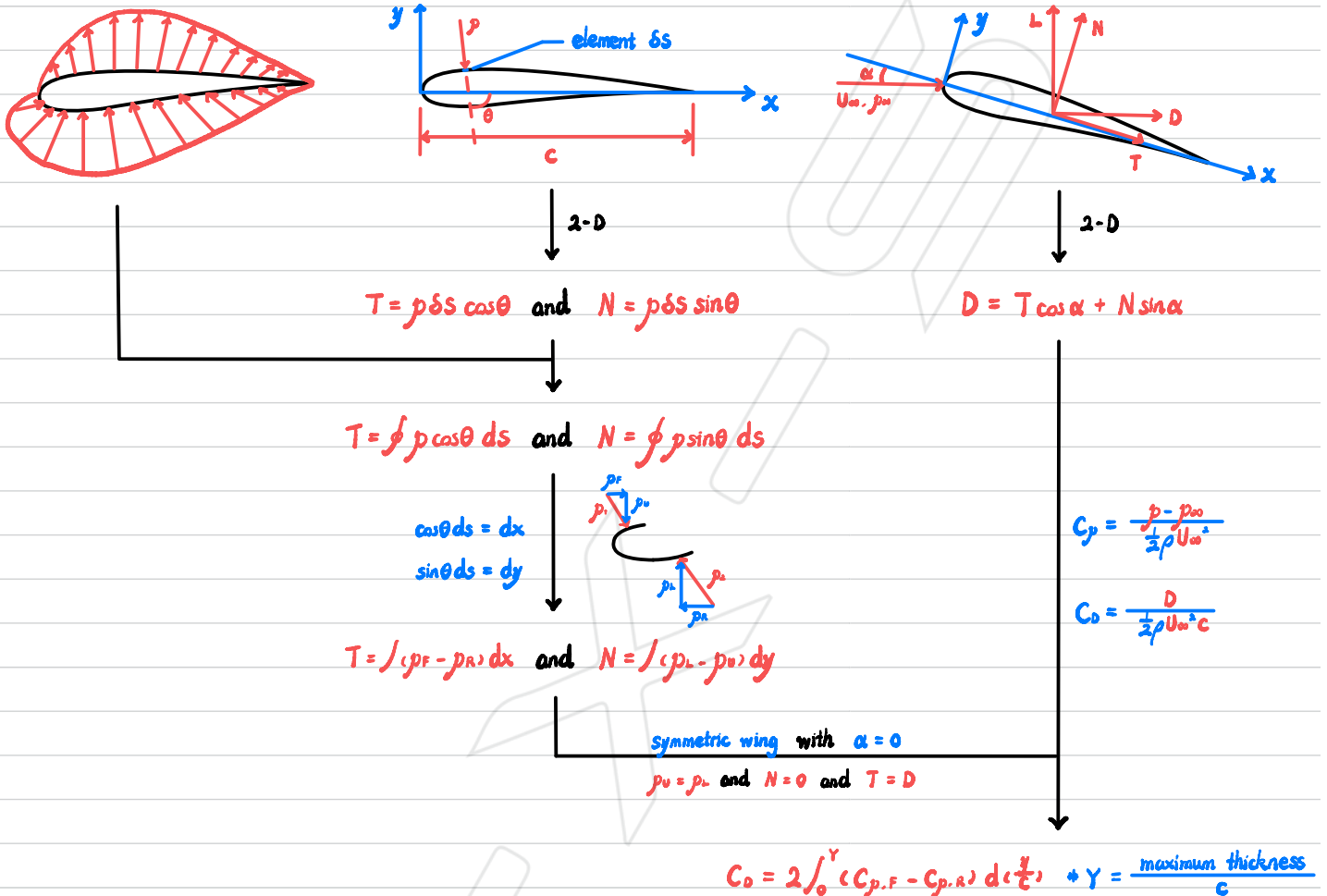


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Profile Drag and Wake Momentum Pre-lab

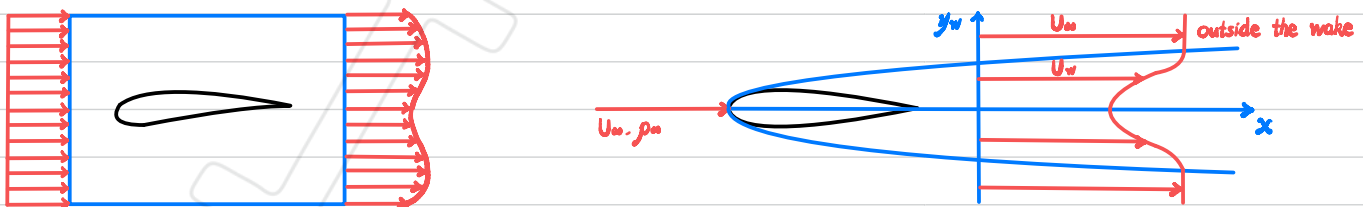
Theory

Pressure (form) drag pressure distribution around airfoil



Skin friction drag

Profile drag $D_{\text{profile}} = D_p + D_{\text{skin friction}}$ momentum deficit in the wake



for steady flow $\xrightarrow{\text{C.V.}}$ 0 + net momentum out of C.V. = 0 + $\rho \int U_w (U_w - U_\infty) dy_w = -D$

$$C_D = \frac{D}{\frac{1}{2} \rho U_\infty^2 c} \rightarrow C_D = \frac{-\rho \int U_w (U_w - U_\infty) dy_w}{\frac{1}{2} \rho U_\infty^2 c} = \frac{2}{c} \int \frac{U_w}{U_\infty} \left(1 - \frac{U_w}{U_\infty}\right) dy_w$$

$$\text{total pressures } H_w \text{ and } H_\infty \rightarrow C_D = \frac{2}{c} \int \left(\frac{H_w - p_\infty}{H_\infty - p_\infty} \right)^{\frac{1}{2}} \left[1 - \left(\frac{H_w - p_\infty}{H_\infty - p_\infty} \right)^{\frac{1}{2}} \right] dy_w$$

$p = p + \frac{1}{2} \rho U^2$ where $p = p_\infty$

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Profile Drag and Wake Momentum

Pre-lab

Procedure

Apparatus

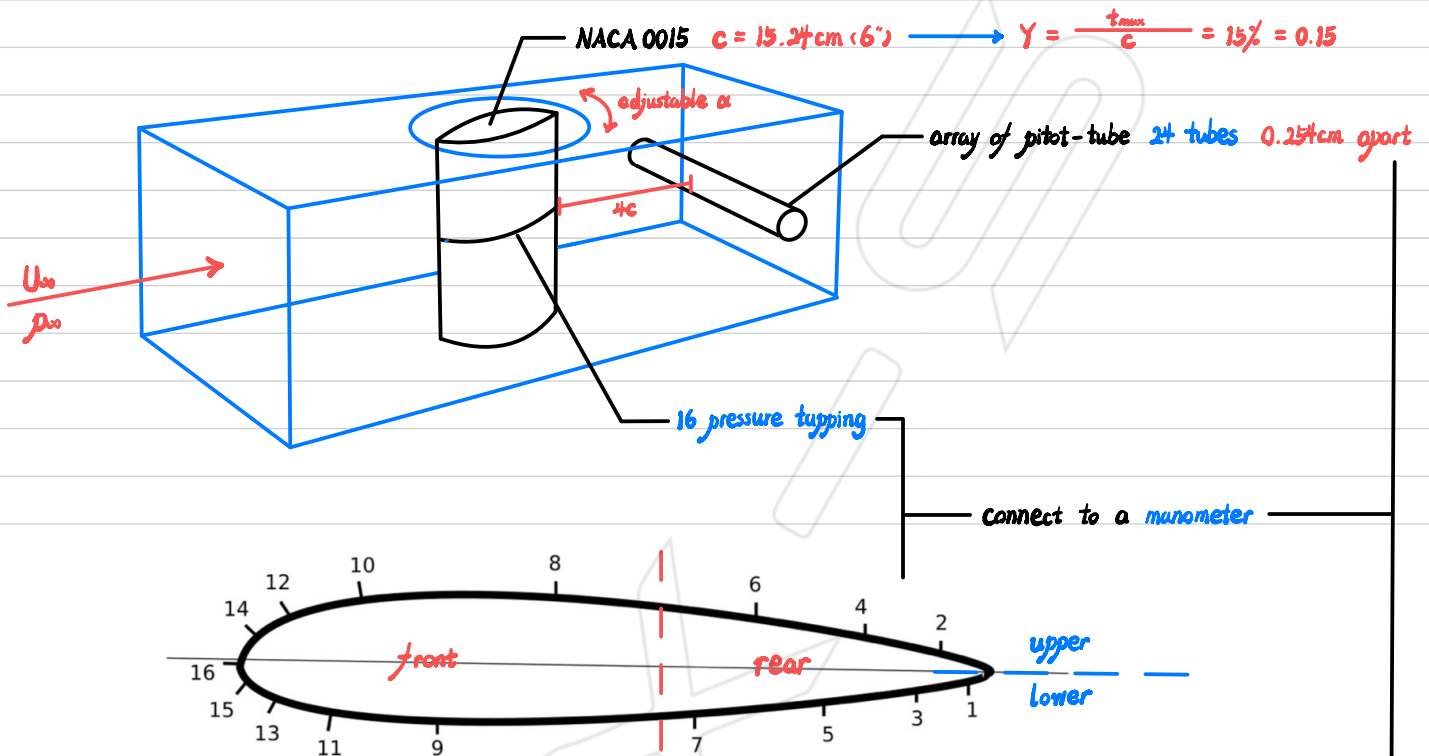


Figure 3: Location of pressure tapings

Hole No.	1	3	5	7	9	11	13	15
x/c	0.958	0.858	0.713	0.525	0.152	0.065	0.020	0.003
y/c	0.0094	0.02815	0.0469	0.06565	0.06565	0.0469	0.02815	0.0094
Hole No.	16	14	12	10	8	6	4	2
x/c	0	0.0087	0.028	0.090	0.303	0.593	0.742	0.888
y/c	0	0.01875	0.0375	0.05625	0.075	0.05625	0.0375	0.01875

Table 1: Surface Pressure Tappings

Tube No.	1	3	5	7	8	9	10	11	12
Distance from Tube 1 [cm]	0	0.508	1.016	1.524	1.778	2.032	2.286	2.54	2.794
Tube No.	13	14	15	16	17	18	20	22	24
Distance from Tube 1 [cm]	3.048	3.302	3.556	3.81	4.064	4.318	4.826	5.334	5.842

Table 2: Wake Pitot Tube Array

Procedure

① $\alpha = 0^\circ$, $\theta_{manometer} = 0^\circ$ $\xrightarrow{U_\infty \uparrow}$ $h_{manometer} = 55 \text{ mm WC}$

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Profile Drag and Wake Momentum Pre-lab

② record data

③ plot h to see if it is symmetric

Error Analysis

Theory

Assume independent variable yields a common formula

$$s_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 s_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 s_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 s_z^2 + \dots} \quad \text{where } s \text{ represent standard deviation}$$

for absolute error

$$\Delta e_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \Delta e_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \Delta e_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 \Delta e_z^2 + \dots}$$

Code provided by

```
function summation = ErrorPropagation(f,variables,values,errs)
```

```
end
```

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Profile Drag and Wake Momentum During lab

Procedure

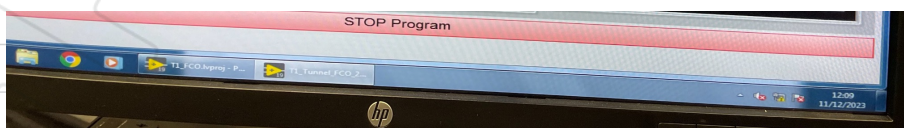
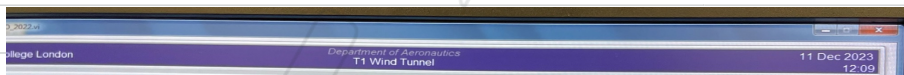
Checklist

- ① pressure tapping align with wake pitot tube array
- ② $\alpha = 0^\circ$, $\theta_{\text{manometer}} = 0^\circ \xrightarrow{U_\infty \uparrow} h_{\text{manometer}} = 55 \text{ mm WC}$
- ③ change manometer angle each time
- ④ plot manometer reading to make sure it is symmetric $\longrightarrow$ change if it is not
- ⑤ find source of error

Data recording

Computer reading

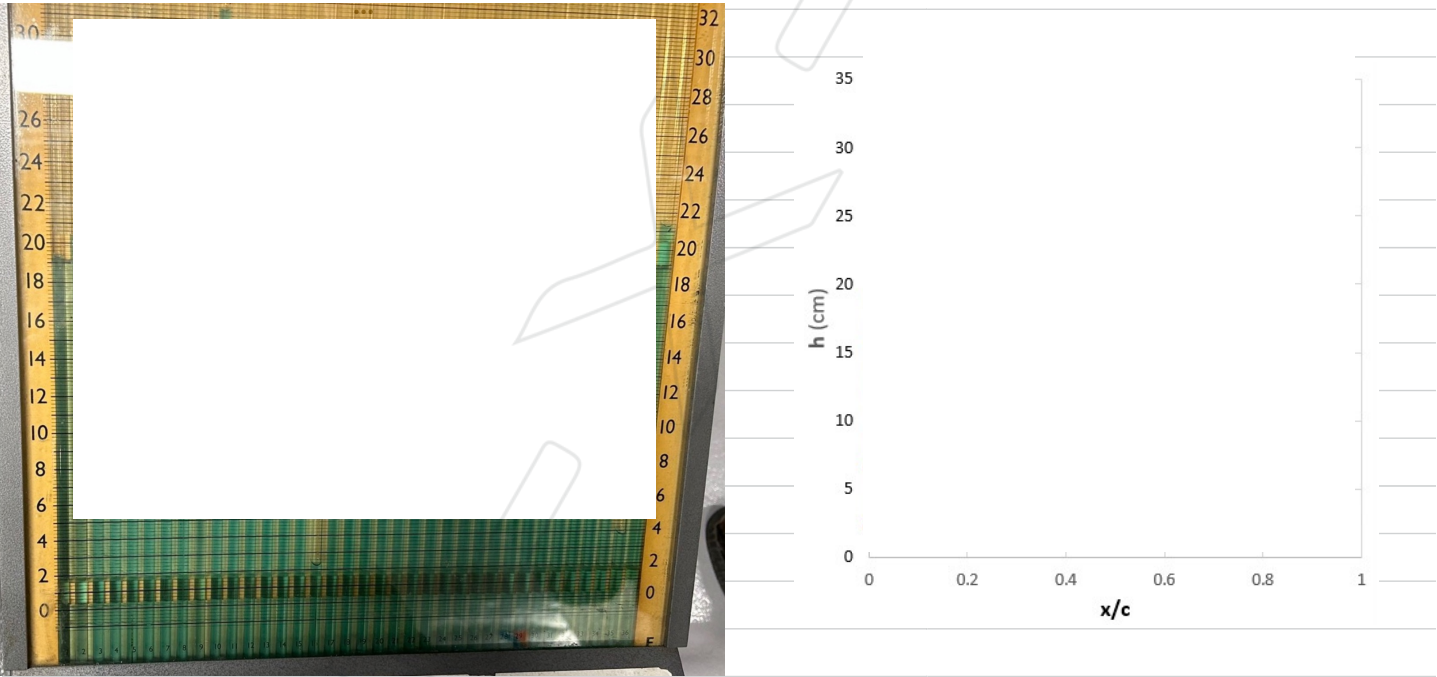
Machine	U_∞ (m/s)	ρ_{alcohol} (kg/m ³)	T (°C)	T (K)	$P_{\text{atmosphere}}$ (kPa)	h_{dynamic} (mm H ₂ O)
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Machine	θ_{reading} ($\pm^\circ$)	$\Delta\theta$ ($\pm \text{rad}$)	T_{reading} ($\pm^\circ\text{C}$)	$T_{\text{experiment}}$ ($\pm^\circ\text{C}$)	ΔT_{total} ($\pm^\circ\text{C}$)	$\Delta h_{\text{dynamic}}$ ($\pm \text{mm H}_2\text{O}$)	$\Delta h_{\text{dynamic}}$ ($\pm \text{m}$)
---------	--	--	---	--	--	--	---

Wing

Manometer	θ ($^{\circ}$)	θ (rad)	h (cm)	h (m)	x/c	y/c
Surface Pressure Tapping						



Manometer	h_{reading} ($\pm$ cm)	$h_{\text{experiment}}$ ($\pm$ cm)	Δh_{total} ($\pm$ cm)	Δh ($\pm$ m)
Surface Pressure Tapping	1			
	2			
	3			
	4			
	5			
	6			
	7			
	8			
	9			
	10			
	11			
	12			
	13			
	14			
	15			
	16			

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Wake

Manometer		θ (°)	θ (rad)	h (cm)	h (m)	x/c	y/c
Surface Pressure Tapping	1						
	2						
	3						
	4						
	5						
	6						
	7						
	8						
	9						
	10						
	11						
	12						
	13						
	14						
	15						
	16						
	Static						
	Total						



Manometer		h_{loading} ($\pm$ cm)	$h_{\text{experiment}}$ ($\pm$ cm)	Δh_{total} ($\pm$ cm)	Δh ($\pm$ m)
Wake Pitot Tube Array	1				
	3				
	5				
	7				
	8				
	9				
	10				
	11				
	12				
	13				
	14				
	15				
	16				
	17				
	18				
	20				
	22				
	24				

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Pressure drag

Profile drag

$$\therefore C_0 = \frac{2}{c} \sum \left(\frac{H_W - p_{00}}{H_{00} - p_{00}} \right)^{\frac{1}{2}} \left[1 - \left(\frac{H_W - p_{00}}{H_{00} - p_{00}} \right)^{\frac{1}{2}} \right] d$$

$$C_{of} = \sqrt{C_o^2 - C_{op}^2}$$

Pressure drag and profile drag integral calculation

[illegible]

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Profile Drag and Wake Momentum Post-lab

C_p vs $\frac{x}{c}$



Figure 6. $-C_p$ v.s. x/c

C_p vs $\frac{y}{c}$



Figure 7. $-C_p$ v.s. y/c

Integrand vs y_w

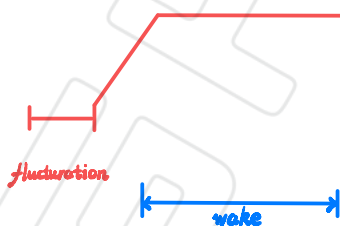


Figure 8. $\left(\frac{H_W - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}} \left(1 - \left(\frac{H_W - p_\infty}{H_\infty - p_\infty}\right)^{\frac{1}{2}}\right)$ vs y_w



Figure 9. Bent pitot tube in wake array

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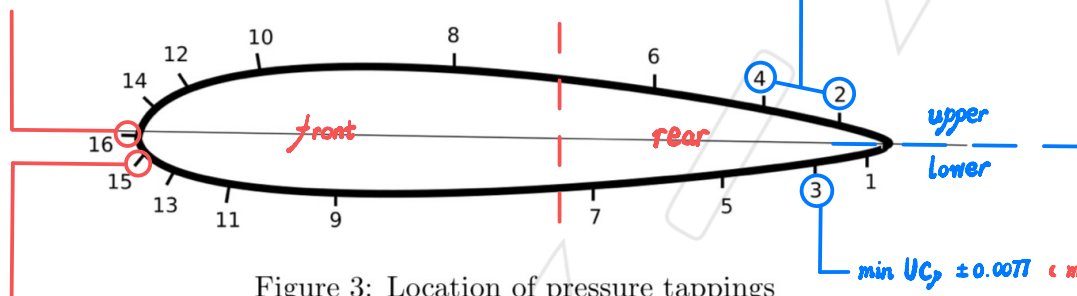
Profile Drag and Wake Momentum Post-lab

Results

		C_D Profile	C_D Pressure	C_D Skin Friction	C_L
Experiment	result				
	error				
CFD					

CAD model not smooth

$$\max UC_p = \pm 0.0454 \quad (\min \% UC_p = 4.38\%)$$



$$\min UC_p = \pm 0.008 \quad (\max \% UC_p = 13.56\%)$$

$$\min UC_p = \pm 0.0077 \quad (\max \% UC_p = 35.62\%)$$

$$\max UC_p = \pm 0.0077 \quad (\max \% UC_p = 4.43\%)$$

Hole No.	1	3	5	7	9	11	13	15
x/c	0.958	0.858	0.713	0.525	0.152	0.065	0.020	0.003
y/c	0.0094	0.02815	0.0469	0.06565	0.06565	0.0469	0.02815	0.0094
Hole No.	16	14	12	10	8	6	4	2
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Distance from Tube 1 [cm]	3.048	3.302	3.556	3.81	4.064	4.318	4.826	5.334	5.842

Table 2: Wake Pitot Tube Array

$$\max UC_{p0} = \pm 0.0144 \quad (\max \% UC_{p0} = 95.45\%)$$

Blockage effects

$$\text{blockage ratio} = \frac{A_{\text{airfoil}}}{A_{\text{wind-tunnel}}} \times 100\% = \frac{0.0113 \text{ m}^2}{0.451 \text{ m}^2} \times 100\% = 2.51\% < 5\% \rightarrow \text{insignificant}$$

Possible Error Sources

- non-uniform surface roughness $\rightarrow L \rightarrow$ lift induced drag $\rightarrow C_{D,p} \uparrow$
- change $T \rightarrow$ change μ and ρ_{air}
- pitot-tube not aligned
- ground not leveled
- parallax error when read manometer